

NHANES

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1 Introduction

NHANES Dataset

- We have 135,310 participants and 5,078 variables
- High dimensional data

1.1 Machine Learning Strategy: Preprocessing with R

1. Upload dataset
2. Check for missing data (replace/delete)
3. Check data types
4. Check if there are any duplicates
5. Check distribution of features and highlight outliers
6. Describe the dataset (mean, median, variance, standard deviation)
 - Plots (distributions, etc.)
 - Check NHANES cycles (different features between cycles)
7. Delete unclean data if not necessary

1.2 Targets

Question: Can chemical, demographic, and metabolic data predict diabetes/hypertension?

- Diabetes, hypertension
- Check for class imbalance
- Sampling/undersampling (SMOTE) if needed, but on the training set only, to avoid data leakage

1.3 Feature Selection (Python)

- Statistical tests of variance
- Correlation
- Mutual information
- Wrapper methods
- Check for collinearity

- Multiple testing correction (FDR, Benjamini–Hochberg)

Features that could be good predictors: Age, BMI, alcohol use, smoking, lifestyle, socio-economic class, caloric intake, obesity, HbA1c, family history, cholesterol, co-morbidities, medication.

- Check gender bias

1.3.1 Modelisation (Python)

Try different models: Logistic Regression, Random Forests, XGBoost, SVM.

- Scale data if needed
- K-stratified cross-validation (5x5), external validation source if available
- Calculate AUC, F1, PR-AUC, Recall, MCC
- Confusion matrix, ROC curve
- SHAP values (plot)
- Compare all model performances
- Check for reproducibility (share models between team members and run multiple times)